

The sandpile group of a de Bruijn graph splits along its repeats

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23 September 2026

Abstract

Two companion papers showed that the sandpile group $\mathcal{K}(G_k)$ of a genome's de Bruijn graph records repeat length, multiplicity and interleaving — a single repeat of length L and multiplicity r contributes $(\mathbb{Z}/r)^{L-k}$, two interleaved repeats add a $\mathbb{Z}/2$, three can add a $\mathbb{Z}/3$ — and showed that the correction is not a rank of the chord diagram. Here we prove the general statement. *Bundle-chain lemma*: in any balanced multidigraph, a chain $v_1 \rightarrow \cdots \rightarrow v_s$ in which every vertex sends all r of its arcs to the next can be contracted to a point, and the sandpile group splits off a direct summand $(\mathbb{Z}/r)^{s-1}$ exactly. The proof is a unimodular change of variables and takes ten lines; the $r = 1$ case is the familiar unitig contraction. Contracting every bundle chain leaves a *skeleton* with one vertex per repeat family and one arc per adjacency of repeat occurrences around the genome — a line-digraph dual of the repeat graph of Pevzner, Tang and Tesler — and

$$\mathcal{K}(G_k) \cong \bigoplus_{\text{chains } c} (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk}).$$

The skeleton is canonical, and the repeat graph itself is the case in which every chain has length two. The first summand is what longer k -mers remove; the second, the sandpile group of the repeat-adjacency skeleton, is what no k removes. For repeats of multiplicity two the second summand is the critical group of the one-vertex map whose edges are the repeats: it is $\text{coker}(A + I)$ for Bouchet's signed interlace matrix A (Merino–Moffatt–Noble), its order is the interlace polynomial at 1 (Arratia–Bollobás–Sorkin), and — as their example shows and we recompute — it is not a function of the unsigned chord-crossing graph, although it happens to be for up to five repeats. For φX174 the decomposition is verified against direct computation at $k = 10, 11, 12$ and extends the computable spectrum to $k = 9$, where the compacted graph (146 branch vertices) exceeded our exact Smith-form range but its skeleton does not: $\mathcal{K}(G_9) = (\mathbb{Z}/2)^{42} \oplus (\mathbb{Z}/4)^4 \oplus \mathbb{Z}/7 \oplus \cdots \oplus \mathbb{Z}/2707831823$, of which $(\mathbb{Z}/2)^{36}$ is length-driven and the rest is the arrangement of repeats. Seven checks, all exact.

1 Introduction

For a circular sequence S and $k \geq 1$, $G_k(S)$ is the de Bruijn multigraph whose vertices are the distinct k -mers and whose arcs are the occurrences of $(k+1)$ -mers; S is an Eulerian circuit, and by the BEST theorem [1, 19] the number of reconstructions of S from its k -mer spectrum is $|\mathcal{K}(G_k)| \prod_v (d^+(v) - 1)!$ with $\mathcal{K}(G_k)$ the sandpile group. The first companion paper [4] showed that $\mathcal{K}(G_k)$ is not determined by the branch statistics and computed it in three model situations:

a single repeat of length L and multiplicity r gives $(\mathbb{Z}/r)^{L-k}$; two repeats of multiplicity two give $(\mathbb{Z}/2)^{2s-2}$ if they are disjoint or nested and $(\mathbb{Z}/2)^{2s-1}$ if interleaved, $s = L - k + 1$; three repeats give a correction that is $\mathbb{Z}/2$, $\mathbb{Z}/3$ or $(\mathbb{Z}/2)^2$ according to their crossing pattern; since the three non-trivial patterns have intersection matrices of the same rank over $\mathbb{F}_2$, the correction is not a function of that rank (Remark 10 sharpens this). The second paper [5] used the group as a coordinate system on reconstructions and found that for φ X174 at $k = 12$ the single surviving $\mathbb{Z}/2$ is a pair of interleaved 12-mers.

Those three model computations are special cases of one structural fact, which this note proves. The fact is local and holds in any balanced multidigraph; its consequence for genomes is that the sandpile group splits as a direct sum of a *length-driven* part, one cyclic factor per shared k -mer of each repeat, and an *arrangement* part, the sandpile group of a repeat-adjacency skeleton.

What is known. Contracting a vertex of in- and out-degree one leaves the sandpile group unchanged [4, Lem. 4]; this is the $r = 1$ case of Lemma 2 and is the reason assemblers' compacted (unitig) graphs [6, 8] carry the same group. For *undirected* graphs with multiple edges the sandpile groups of thick trees [3] and thick cycles [2] are known, and the thick-tree answer, $\bigoplus \mathbb{Z}/m_e$ over the edge multiplicities, looks like our formula; but the undirected analogue of Lemma 2 is false (Remark 4), so the lemma is not a corollary of those results. The nearest directed relative is Levine's theorem that $\mathcal{K}(G)$ is a quotient of $\mathcal{K}(LG)$ for the line digraph [10, 16]. The Eulerian-path formulation of assembly is due to Pevzner, Tang and Waterman [18], and the *repeat graph*, in which repeat families are edges of multiplicity > 1 and vertices are repeat boundaries, to Pevzner, Tang and Tesler [17]; it is the central object of the long-read assembler Flye [9]. Our skeleton has one *vertex* per repeat family and one arc per adjacency of occurrences, so it stands to the repeat graph roughly as a line digraph to its graph; we keep the two apart by name. For two-copy repeats the skeleton is the directed medial graph of a one-vertex map, and its sandpile group is the critical group of that map in the sense of Merino, Moffatt and Noble [13], which is where the question of what determines it has already been answered (Remark 10). We have not found the $r > 1$ splitting of Lemma 2 in the sandpile literature, nor the sandpile group of the repeat graph considered as such; a search cannot establish a negative.

2 Setting

All digraphs are finite multidigraphs, balanced ($d^-(v) = d^+(v)$ at every vertex) and connected. The out-Laplacian is $\Delta = D^+ - A$; the sandpile group is $\mathcal{K}(G) = \text{coker}(\Delta_s^T)$, the row and column of a sink s deleted; for balanced G the isomorphism class does not depend on s [7, Lem. 4.12], and $|\mathcal{K}(G)|$ is the number of spanning arborescences toward s . We compare finite abelian groups by their elementary divisors.

Definition 1 (Bundle chain). A *bundle chain* of multiplicity r and length $s \geq 2$ is a sequence of distinct vertices $v_1, \dots, v_s$ such that for each $t < s$ the vertex v_t has out-degree r and all r of its out-arcs go to v_{t+1} , and v_s has out-degree r . Since G is balanced, every in-arc of v_{t+1} ($t < s$) comes from v_t . The *contraction* G/C replaces the chain by a single vertex carrying the in-arcs of v_1 and the out-arcs of v_s ; it is again balanced, and an arc $v_s \rightarrow v_1$ becomes a loop, which does not affect Δ .

3 The lemma

Lemma 2 (Bundle-chain contraction). *Let $C = (v_1, \dots, v_s)$ be a bundle chain of multiplicity r in a balanced multidigraph G with at least one vertex outside C . Then*

$$\mathcal{K}(G) \cong (\mathbb{Z}/r)^{s-1} \oplus \mathcal{K}(G/C).$$

Proof. Root at a vertex outside C and write x_v for the generator of the cokernel presentation at each non-root vertex, the relations being the rows of Δ . Change generators on the chain:

$$z := x_{v_1}, \quad y_t := x_{v_t} - x_{v_{t+1}} \quad (1 \leq t < s),$$

so that $x_{v_t} = z - \sum_{u < t} y_u$ and in particular $x_{v_s} = z - \sum_{u < s} y_u$; this is a unimodular (triangular) change of basis. The rows of Δ fall into three kinds.

- The row of v_t for $t < s$ is $r x_{v_t} - r x_{v_{t+1}} = r y_t$.
- The row of v_s is $r x_{v_s} - \sum_w c_{v_s w} x_w = r z - r \sum_{u < s} y_u - \sum_w c_{v_s w} x_w$. Subtracting the rows $r y_u$ — a unimodular row operation — leaves $r z - \sum_w c_{v_s w} x_w$, which is the row of the contracted vertex in $\Delta(G/C)$.
- The row of a vertex $u \notin C$ involves x_{v_t} only through arcs from u into the chain, and all of those land on v_1 ; so it reads $d^+(u)x_u - \sum_w c_{uw} x_w$ with $x_{v_1} = z$, which is its row in $\Delta(G/C)$.

The relation lattice is therefore generated by $\{r y_t\}_{t < s}$ together with the rows of $\Delta(G/C)$ in the variables z and x_u , and the generators separate accordingly. Hence $\mathcal{K}(G) \cong \bigoplus_{t < s} \mathbb{Z}/r \oplus \mathcal{K}(G/C)$. $\square$

If G consists of the chain alone — the only vertices are $v_1, \dots, v_s$ and the r arcs of v_s return to v_1 — the same computation rooted at v_1 gives $\mathcal{K}(G) \cong (\mathbb{Z}/r)^{s-1}$; this is the single-repeat theorem of [4, Thm. 5].

Remark 3. For $r = 1$ the summand is trivial and the lemma says that smoothing a vertex of in- and out-degree one preserves the group, which is [4, Lem. 4]. The content of the lemma for $r > 1$ is that a vertex which *cannot* be smoothed — it has r parallel exits — can nevertheless be removed at the exact cost of a $\mathbb{Z}/r$.

Remark 4 (The direction is essential). The undirected statement one might guess — a path of doubled edges contributes $(\mathbb{Z}/2)^{\text{length}-1}$ — is false. The undirected 4-cycle with every edge doubled has reduced Laplacian $2M$ with M the reduced Laplacian of the simple 4-cycle, whose Smith form is $\text{diag}(1, 1, 4)$; so its sandpile group is $(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/8$, whereas the directed doubled 4-cycle is a single bundle chain and has $(\mathbb{Z}/2)^3$. What makes the directed lemma work is that every arc *into* the chain lands on its head v_1 and every arc *out* of it leaves from v_s with the full multiplicity r ; an undirected doubled edge carries flow both ways and the change of variables in the proof no longer separates.

4 The splitting theorem

Contracting bundle chains one after another terminates, since each contraction reduces the number of vertices, and the result contains no bundle chain of length ≥ 2 . We call it the *skeleton* $\text{Sk}(G)$. Its sandpile group is well defined by the lemma; the skeleton itself is too.

Lemma 5 (The skeleton is canonical). *Call $v \rightarrow w$ a chain step if $v \neq w$, all out-arcs of v go to w , and $d^+(w) = d^+(v)$. In a balanced multidigraph the chain steps form a partial injection on the vertex set, so the maximal bundle chains are its maximal paths: they are pairwise vertex-disjoint and determined by G . Contracting them in any order gives the same digraph up to isomorphism, which is chain-free.*

Proof. If $v \rightarrow w$ is a chain step then w receives $d^+(v) = d^+(w) = d^-(w)$ arcs from v , hence all of its in-arcs, so no $v' \neq v$ steps to w ; and v has a single target. Thus the step relation is a partial injection and its maximal paths partition the vertices it touches (a path may close into a cycle only if G is a single closed chain, when the skeleton is one vertex with r loops). Contracting a maximal chain C to a vertex c changes no degree outside C , gives c the in-arcs of v_1 and the out-arcs of v_s , and so creates no new chain step: $u \rightarrow c$ would require $u \rightarrow v_1$ to have been a step and $c \rightarrow w$ would require $v_s \rightarrow w$ to have been one, both excluded by maximality. Nor does it destroy a step outside C . Hence the contractions of distinct maximal chains commute, and the result is G with every maximal chain contracted. $\square$

Check (E) verifies this on 100 random balanced digraphs with planted chains: the partition into maximal chains, the multiset of (r_c, ℓ_c) and the skeleton's group are invariant under random relabelling, and the skeleton contains no chain step.

Theorem 6 (Splitting). *For any balanced connected multidigraph G , with $\{(r_c, \ell_c)\}$ the multiplicities and lengths of the chains contracted on the way to a skeleton,*

$$\mathcal{K}(G) \cong \bigoplus_c (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk}(G)), \quad |\mathcal{K}(G)| = \prod_c r_c^{\ell_c-1} \cdot |\mathcal{K}(\text{Sk}(G))|.$$

Proof. Induction on the number of contractions, using Lemma 2 at each step. $\square$

Now let S be a sequence with repeats $W_1, \dots, W_m$ of lengths L_i and multiplicities r_i , and suppose the *clean* hypothesis of [4]: the k -mers inside W_i occur exactly r_i times each and every other k -mer once. Write w for the circular word of repeat occurrences and D_w for the *connector digraph*: one vertex per repeat, one arc $i \rightarrow j$ for each occurrence of W_i immediately followed around the circle by an occurrence of W_j (loops deleted). D_w is balanced with out-degree r_i at vertex i .

Corollary 7 (Clean sequences). *Under the clean hypothesis, with $s_i = L_i - k + 1$,*

$$\mathcal{K}(G_k(S)) \cong \bigoplus_{i=1}^m (\mathbb{Z}/r_i)^{s_i-1} \oplus \mathcal{K}(D_w).$$

Proof. After unitig contraction the graph consists of the m spines — bundle chains of multiplicity r_i and length s_i — joined by the connector arcs; contracting the spines gives D_w . Apply Theorem 6. $\square$

Corollary 8 (The published cases). *For $m = 1$, D_w is a single vertex and $\mathcal{K} = (\mathbb{Z}/r)^{s-1}$ [4, Thm. 5]. For $m = 2$, $r_1 = r_2 = 2$: D_w is trivial for $AAB B$ and $ABBA$ and is the doubled 2-cycle, with $\mathcal{K}(D_w) = \mathbb{Z}/2$, for $ABAB$ [4, Thm. 6]. For $m = 3$ the four crossing patterns give $\mathcal{K}(D_w) = 0, \mathbb{Z}/2, \mathbb{Z}/3, (\mathbb{Z}/2)^2$ [4, Prop. 8]; the $\mathbb{Z}/3$ is the sandpile group of the complete symmetric digraph on three vertices, which is D_w for $ABACBC$.*

Remark 9 (The skeleton and the repeat graph). The duality with the repeat graph of [17, 9] can be made exact. Under the clean hypothesis, collapse each spine of the compacted graph to a single arc: the result is the multidigraph RG_w with two vertices h_i, t_i per family, r_i parallel arcs $h_i \rightarrow t_i$, and one arc $t_i \rightarrow h_j$ per adjacency ij in w — repeats as multi-arcs between their boundary vertices, unique segments as single arcs, which is the repeat graph. Each pair (h_i, t_i) is a bundle chain of length two, and contracting it identifies the multi-arc with a vertex: D_w is the transition digraph of the repeat arcs of RG_w (the line digraph, restricted to repeat arcs, with unique segments as the adjacencies). By Lemma 2,

$$\mathcal{K}(\text{RG}_w) \cong \bigoplus_{i=1}^m \mathbb{Z}/r_i \oplus \mathcal{K}(D_w),$$

so the sandpile group of the repeat graph is the arrangement part plus one $\mathbb{Z}/r_i$ per family — the length-driven part at $s_i = 2$ (check (E), 200 random words). The two objects carry the same information; we work with D_w because it has no length-driven residue.

Remark 10 (Which invariant of the chord diagram). For $r_i = 2$ the $2m$ occurrences form a chord diagram with m chords — a double-occurrence word, or a one-vertex map (bouquet) with the repeats as edges — and D_w is a 2-in 2-out digraph. On what invariant of the diagram does $\mathcal{K}(D_w)$ depend? Not on the rank of the unsigned intersection matrix, over $\mathbb{F}_2$ or over $\mathbb{Z}$, and not on its integer Smith form: for $m = 3$ the three non-trivial patterns all have $\mathbb{F}_2$ -rank 2, and $A A B C B C$ (one crossing) and $A B A C B C$ (path) share the Smith form $(1, 1, 0)$ while their groups are $\mathbb{Z}/2$ and $\mathbb{Z}/3$; for $m = 5$ the rank-2 diagrams carry nine different groups, among them $\mathbb{Z}/7$ and $\mathbb{Z}/9$ (check (D)). The order $|\mathcal{K}(D_w)|$ is the number of Eulerian circuits of D_w , and this is a function of the intersection graph H alone (the *interlace graph* of [11], whose Definition 1 is exactly chord crossing): it is the interlace polynomial $q(H; 1)$ [11, Thm. 9, Cor. 13], with roots in Martin’s and Las Vergnas’ circuit-partition polynomials [12]. Exhaustively for $m \leq 5$ the group is also constant on the 4, 11 and 34 isomorphism classes of H — but this is an accident of small m . Merino, Moffatt and Noble [13, Rem. 6.7] give two 6-chord bouquets, 1 2 3 4 5 6 1 3 2 4 6 5 and 1 2 3 6 5 4 1 3 2 5 6 4, with the same interlace graph (K_6 minus two disjoint edges, so $|\mathcal{K}| = 18$ for both) and critical groups $\mathbb{Z}/3 \oplus \mathbb{Z}/6$ and $\mathbb{Z}/18$; check (D) recomputes both from D_w and exhibits two further such pairs. What does determine the group is the *signed* interlace matrix A of Bouchet ($A_{ef} = \pm 1$ according as the occurrences run $efef$ or $fe fe$, 0 if not interlaced): by [13, Thm. 7.3], whose directed medial graph of a bouquet is our D_w ,

$$\mathcal{K}(D_w) \cong \text{coker}(A + I),$$

the critical group of the map; its order $\det(A + I)$ as an Euler-circuit count is older [14, 15]. Check (D) confirms the isomorphism on every double-occurrence word with $m \leq 5$ and on 1200 random words with $6 \leq m \leq 9$. So the arrangement part of an assembly’s ambiguity, for two-copy repeats, is the critical group of the bouquet whose edges are the repeats, computable from a signed $m \times m$ matrix rather than from the de Bruijn graph; and the reason it is not a function of the chord-crossing graph is the reason [13] record: the signs matter. For $r_i > 2$ the occurrences form a hypergraph rather than a chord diagram and we know of no analogue.

Remark 11 (What k can and cannot remove). Increasing k shortens every spine: s_i decreases by one per unit of k , and the length-driven summand $\bigoplus (\mathbb{Z}/r_i)^{s_i-1}$ shrinks accordingly, vanishing at $k = \max L_i$. The skeleton summand $\mathcal{K}(D_w)$ does not depend on k at all, as long as $k \leq \min_i L_i$. So the global ambiguity of an assembly splits into a part that longer k -mers resolve and a part

that only information beyond the repeats — long reads, pairs, a reference — can resolve, and the second part is exactly the sandpile group of the repeat-adjacency skeleton. By BEST applied to D_w , $|\mathcal{K}(D_w)| \cdot \prod_i (r_i - 1)!$ is the number of Eulerian circuits of D_w , i.e. of circular words with the same adjacency counts as w : the number of ways to thread the repeats, ignoring their lengths. That this is $|\mathcal{K}(D_w)|$ up to the local factor is the repeat-level form of the decomposition of [4].

5 φ X174

Table 1 applies the theorem to the first sequenced genome (NC_001422.1, 5386 bp). At each k the compacted graph is bundle-contracted; the chains, the skeleton, and the two summands are listed, and where the compacted graph is small enough the group is also computed directly for comparison. Two things happen. The direct and decomposed groups agree wherever both exist ($k = 10, 11, 12$). And at $k = 9$, where the compacted graph has 146 branch vertices and was beyond the dense exact Smith form used in [4], the skeleton has 110 and is not: the spectrum extends by one level, with the length-driven part $(\mathbb{Z}/2)^{36}$ split off exactly and the arrangement part read off the skeleton.

k	branch	chains	skeleton	$\mathcal{K}(\text{Sk})$ (arrangement part)	length-driven part
12	2	1	1	0	$\mathbb{Z}/2$
11	10	2	8	$\mathbb{Z}/3 \oplus \mathbb{Z}/11$	$(\mathbb{Z}/2)^2$
10	47	8	37	$\mathbb{Z}/4 \oplus \mathbb{Z}/1076338579$	$(\mathbb{Z}/2)^{10}$
9	146	28	110	$(\mathbb{Z}/2)^6 \oplus (\mathbb{Z}/4)^4 \oplus \mathbb{Z}/7 \oplus \mathbb{Z}/523 \oplus \mathbb{Z}/3779$ $\oplus \mathbb{Z}/5297 \oplus \mathbb{Z}/6561 \oplus \mathbb{Z}/7669 \oplus \mathbb{Z}/2707831823$	$(\mathbb{Z}/2)^{36}$
8	401	65	312	(skeleton beyond the exact-SNF range used)	

Table 1: φ X174 under bundle-chain contraction: number of branch vertices of the compacted graph, number of bundle chains contracted, number of skeleton vertices, and the two summands. $\mathcal{K}(G_k)$ is their direct sum; at $k = 10, 11, 12$ this agrees with the group computed directly on the compacted graph (check (F)). All chains have multiplicity 2. All entries exact.

The reading at $k = 11$ is the cleanest: $\mathcal{K}(G_{11}) = (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/11$, and the theorem says which is which — the two $\mathbb{Z}/2$'s are two repeat families each with one extra shared 11-mer beyond the minimum, and $\mathbb{Z}/3 \oplus \mathbb{Z}/11$, of order 33, is the arrangement ambiguity of the eight-vertex skeleton, which no choice of $k \leq 11$ can reduce.

6 Verification

The script `repeat_splitting.py` runs the seven checks of Table 2 in about thirty seconds, all passing; its output is archived as `repeat_splitting_output.txt`. All arithmetic is exact; groups are compared by elementary divisors.

7 Scope

Proved. Lemma 2, Theorem 6 and its corollaries, for arbitrary balanced multidigraphs and arbitrary multiplicities; the clean hypothesis enters only in identifying the skeleton with D_w .

key	check	mode
(A)	Corollary 7 on 120 clean random sequences, $m \leq 4$, $r_i \leq 4$	exact
(B)	Lemma 2 on 60 random balanced digraphs with a planted chain, $r \leq 6$, $s \leq 4$	exact
(C)	Corollary 8: the nine published values are the formula on D_w	exact
(D)	Remark 10: all double-occurrence words, $m \leq 5$; rank, Smith form, graph	exact
(E)	Lemma 5 on 100 relabelled digraphs; Remark 9 on 200 words	exact
(F)	Table 1: decomposition vs. direct computation; extension to $k = 9$	exact

Table 2: The verification battery, 7/7 (check (E) has two rows).

Quoted. Sink-independence [7]; BEST [1, 19]; the Eulerian formulation and the repeat graph [18, 17, 9]; the thick-graph and line-digraph results that are the lemma’s nearest relatives [3, 2, 10, 16]; the special cases and the φ X174 spectrum [4, 5].

Open. (a) A structure theorem for $\mathcal{K}(D_w)$ itself — for which circular words is it trivial, and what bounds its exponent? For $r_i = 2$ this is the critical group of a bouquet (Remark 10), and [13] note that for 3-connected regular delta-matroids the group is well defined on the delta-matroid; whether $\mathcal{K}(D_w)$ is determined by the interlace graph whenever that graph is prime (no split) we have not examined. For $r_i > 2$ nothing of the kind is available. (b) Real genomes are not clean: the skeleton of φ X174 at $k = 9$ has 110 vertices and carries factors as large as 2707831823. All six factors above 100 in Table 1 (523, 3779, 5297, 7669, 1076338579, 2707831823) are prime, which is what the sandpile group of a generic digraph looks like (typically cyclic, or nearly so); we see no reason to expect a topological reading of a large prime cyclic factor and claim none. (c) Double strands: reverse complementation reverses arc direction, so a bidirected version of Lemma 2 must start from the signed Laplacian; we have not attempted it.

Provenance

Unrefereed preprint. Lemmas 2 and 5 are proved in full above and are the only new mathematics; Remark 10 is a translation of published results [11, 13] into the assembly setting, verified computationally — an earlier draft of this note conjectured that the group was an invariant of the unsigned interlace graph, which [13, Rem. 6.7] had already shown to be false; everything else is either quoted or an exact computation reproducible in seconds. We searched for the $r > 1$ splitting in the sandpile literature and for any use of the sandpile group of a repeat graph or repeat-adjacency skeleton and found neither; the undirected thick-graph results [3, 2] are the closest, and Remark 4 shows they do not imply the lemma.

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